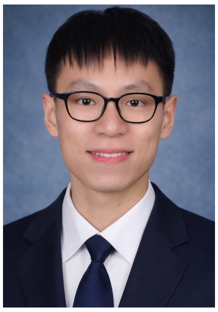
Ziheng Xiao

Energy Research Institute @ NTU (ERI@N), Nanyang Technological University (NTU), Singapore
ziheng.xiao@ntu.edu.sg

Personal Information



Gender	Male
Date of Birth	Mar. 9, 1995
Area of Specialism	Power Electronics, Artificial Intelligence
Web of Science ID	AAA-8268-2020
Scopus ID	57205659683
ORCID ID	0000-0002-4144-3532
Google Scholar	`https://scholar.google.com/citations?user=nABfweAAAAAJ`
Homepage	`https://lumia-xiao.github.io/`

Education and Working Experience

Education

Year	Degree	Discipline	Institution
2017–2022	Doctor of Philosophy (supervised by Professor An Luo, Academician of the Chinese Academy of Engineering)	Electrical Engineering	College of Electrical and Information Engineering, Hunan University
2013–2017	Bachelor	Electrical Engineering and Automation	College of Electrical and Information Engineering, Hunan University

Working Experience

Year	Working Experience
2026–Present	**Senior Research Fellow**, Energy Research Institute @ NTU (ERI@N), supervised by Professor Yi Tang, Associate Chair of the School of Electrical and Electronic Engineering (EEE), NTU
2023–Present	**Part-time Lecturer**, School of Electrical and Electronic Engineering (EEE), NTU; responsible for three postgraduate courses: EE6506 Semiconductor Based Converter in Renewable Energy Systems EE6514 Special Topics in Clean Energy System Design EE6501 Power Electronic Converters
2022–2026	**Postdoctoral Research Fellow**, Energy Research Institute @ NTU (ERI@N), supervised by Professor Yi Tang

Projects Participated

Year	Projects	Direct Costs
2026–2030	Efficiency Improvement Advanced Flow-based Energy Storage Innovations for Scalability & Systemic Industry Collaboration Projects (IAF-ICP), Singapore	**S$ 16.5 M**
2024–2027	Advancing Electrified Transportation and Intelligent Systems through Physics-Informed Machine Learning in Wireless Power Transfer (R24I6IR134), Agency for Science, Technology and Research (A*STAR), Singapore	**S$ 350 k**
2024–2027	Optimizing Wireless Power Transfer Efficiency Through Transfer Learning-Based Precise Power Loss Modelling (RG73/24), Ministry of Education (MOE), Singapore	**S$ 160 k**
2021–2025	Advanced Energy Management and Data Fusion for Smart Homes and Smart Grids, Lite-On Singapore Pte. Ltd., Singapore	**S$ 5 M**
2019–2021	Natural Science Foundation of China under Grant (51807057), China	**¥ 240 k**
2019–2021	Hunan Province Youth Fund Project (2019JJ50038), China	**¥ 50 k**
2018–2023	Project of Megmeet (University-Industry Collaboration Programme), China	**¥ 1.2 M**
2018–2020	Hunan Science and Technology Innovation Plan Project (2017XK2104), China	**¥ 5 M**

Citation Summary

Database	Without Self-Citations	With Self-Citations	H-index
Scopus		**598**	**15**
Web of Science (SCI)	**333**	**436**	**12**
Google Scholar		**677**	**15**

Citation metrics accessed on Jul. 22, 2026.

Publication Summary

Journal Publications

Journal	TIE	TPEL	TTE	JESTPE	Others	Total
First Author	3	15	1	2	2	**23**
Corresponding Author	0	2	1	0	0	**3**
Co-Author	3	8	1	2	1	**15**

Conference Publications

Conference	APEC	ECCE	IECON	ECCE-Asia	Others	Total
First Author	0	2	2	2	3	**9**
Co-Author	6	4	2	4	4	**20**

Journal Papers – First Author

No.	Publication Title	IF
[1]	**Z. Xiao**, Z. He, R. Guan and A. Luo, "Piecewise-Approximated Time Domain Analysis of LLC Resonant Converter Considering Parasitic Capacitors and Deadtime," ***IEEE Trans. Power Electron.***, vol. 38, no. 1, pp. 578–592, Jan. 2023, doi: 10.1109/TPEL.2022.3207329.	6.7
[2]	**Z. Xiao**, Y. Jiang, T. Sun, Y. Wu and Y. Tang, "Refining Power Converter Loss Evaluation: A Transfer Learning Approach," ***IEEE Trans. Power Electron.***, vol. 39, no. 4, pp. 4313–4324, Apr. 2024, doi: 10.1109/TPEL.2023.3349178.	6.7
[3]	**Z. Xiao**, Z. He, H. Wang, A. Luo, Z. Shuai and J. M. Guerrero, "General High-Frequency-Link Analysis and Application of Dual Active Bridge Converters," ***IEEE Trans. Power Electron.***, vol. 35, no. 8, pp. 8673–8688, Aug. 2020, doi: 10.1109/TPEL.2019.2963746.	6.7
[4]	**Z. Xiao**, Z. He, Z. Li, L. Zhu and L. Wang, "Unified Description and Optimization Method of Dual Active Bridge DC–DC Converters," ***IEEE Trans. Power Electron.***, vol. 37, no. 10, pp. 11839–11854, Oct. 2022, doi: 10.1109/TPEL.2022.3177245.	6.7
[5]	**Z. Xiao**, Z. Li, Z. He, H. Wang, Y. Tang and A. Luo, "Computer-Aided Time-Domain Operation and Design Optimization of Resonant Converters," ***IEEE J. Emerg. Sel. Topics Power Electron.***, vol. 11, no. 4, pp. 4164–4177, Aug. 2023, doi: 10.1109/JESTPE.2023.3275637.	5.5
[6]	**Z. Xiao**, Z. He, R. Guan, Z. Li, B. Zhou and A. Luo, "A Three-Terminal Submodule Based High DC Conversion Ratio System With Self-Balance Feature," ***IEEE Trans. Power Electron.***, vol. 37, no. 5, pp. 5650–5663, May 2022, doi: 10.1109/TPEL.2021.3131795.	6.7
[7]	**Z. Xiao**, Y. Jiang, F. Deng, Z. Yao and Y. Tang, "A Data-Driven Control Parameters Optimization Method for Dual Active Bridge Converters," ***IEEE Trans. Ind. Electron.***, vol. 71, no. 11, pp. 14054–14066, Nov. 2024, doi: 10.1109/TIE.2024.3370950.	7.7
[8]	**Z. Xiao** et al., "A Hybrid Data-Driven Power Loss Minimization Method of Dual-Active Bridge Converters," ***IEEE Trans. Power Electron.***, vol. 39, no. 5, pp. 5820–5832, May 2024, doi: 10.1109/TPEL.2024.3368330.	6.7
[9]	**Z. Xiao**, Y. Zeng, and Y. Tang, "Swift and Seamless Start-Up of DAB Converters in Constant and Variable Frequency Modes," ***IEEE Trans. Ind. Electron.***, vol. 72, no. 5, pp. 4763–4775, May 2025, doi: 10.1109/TIE.2024.3481900.	7.7
[10]	**Z. Xiao** et al., "Analysis and Design of a Modular Self-Balance High-Voltage Input DC–DC Topology," ***IEEE J. Emerg. Sel. Topics Power Electron.***, vol. 10, no. 2, pp. 2276–2289, Apr. 2022, doi: 10.1109/JESTPE.2022.3140898.	5.5
[11]	**Z. Xiao**, X. Li and Y. Tang, "A Lightweight Artificial Neural Network Start-Up Controller for CLLC Resonant Converters," ***IEEE Trans. Power Electron.***, vol. 39, no. 11, pp. 14775–14786, Nov. 2024, doi: 10.1109/TPEL.2024.3436847.	6.7
[12]	**Z. Xiao**, Z. Yao, F. Deng, L. Zhang and Y. Tang, "Seamless Rotational Phase Shedding for Multiphase DC–DC Converters," ***IEEE Trans. Power Electron.***, vol. 39, no. 5, pp. 4996–5001, May 2024, doi: 10.1109/TPEL.2024.3359604.	6.7

No.	Publication Title	IF
[13]	**Z. Xiao**, M. Xiao, Z. He, L. Wang, Z. Li and H. Wang, "Light Load Efficiency Improvement of DAB Converters With Optimal Switching Sequences," ***IEEE Trans. Power Electron.***, vol. 40, no. 7, pp. 9343–9356, Jul. 2025, doi: 10.1109/TPEL.2024.3513412.	6.7
[14]	**Z. Xiao** et al., "Exploring Optimal Switching Patterns for Rapid Start-Up in Synchronous Boost Converters," ***IEEE Trans. Ind. Electron.***, vol. 72, no. 8, pp. 7897–7909, Aug. 2025, doi: 10.1109/TIE.2024.3522511.	7.7
[15]	**Z. Xiao**, Z. Yao, F. Deng, Y. Zhang and Y. Tang, "Exploring the Extreme Dynamic Transition Response of Dual Active Bridge Converters," ***IEEE Trans. Power Electron.***, vol. 40, no. 5, pp. 6988–7001, May 2025, doi: 10.1109/TPEL.2025.3532400.	6.7
[16]	**Z. Xiao**, F. Deng, Z. Yao and Y. Tang, "Soft-Morphing: Unlocking the Potential of CLLC Converters for Ultrawide Range Applications," ***IEEE Trans. Power Electron.***, vol. 40, no. 1, pp. 1289–1304, Jan. 2025, doi: 10.1109/TPEL.2024.3480909.	6.7
[17]	**Z. Xiao**, J. Wu, Z. He, Z. Yao and Y. Tang, "Optimal Switching Pattern Control of the CLLC Resonant Converters in Dynamic Processes," ***IEEE Trans. Power Electron.***, vol. 39, no. 12, pp. 16268–16282, Dec. 2024, doi: 10.1109/TPEL.2024.3457899.	6.7
[18]	**Z. Xiao**, F. Deng, Z. Yao and Y. Tang, "Minimizing and Balancing Power Losses in Multiphase Paralleled Bridge Legs," ***IEEE Trans. Power Electron.***, vol. 40, no. 9, pp. 13835–13851, Sept. 2025, doi: 10.1109/TPEL.2025.3569667.	6.7
[19]	**Z. Xiao**, Z. He, L. Zhang, Z. Yao, and Y. Tang, "Resonant DC Transformer with Integer Multiple Frequency for High Step-Ratio Application," ***IEEE Trans. Transport. Electrific.***, vol. 12, no. 1, pp. 1416–1429, 2026, doi: 10.1109/TTE.2025.3627551.	7.2
[20]	**Z. Xiao**, T. Sun, Z. Yao and Y. Tang, "Optimizing Multi-phase DC-DC Converters via Inter-leaving/Intra-leaving and Phase Shedding," ***IEEE Trans. Power Electron.***, vol. 41, no. 4, pp. 6630–6646, 2026, doi: 10.1109/TPEL.2025.3631793.	6.7
[21]	**Z. Xiao**, Z. He, Y. Ning, H. Wang, A. Luo, Y. Chen and J. Chen, "Optimization of LLC Resonant Converter With Two Degrees of Freedom Based on Operation Stage Trajectory Analysis," ***IEEE Access***, vol. 9, pp. 79629–79642, 2021.	
[22]	**Z. Xiao**, L. Zhang, Y. Zeng, F. Deng, Z. Yao and Y. Tang, "L-MMC: A Building Block for High-Efficiency and High-Power-Density DC–DC Conversion," ***IEEE Trans. Power Electron.***, 2026.	6.7
[23]	**Z. Xiao**, Z. He, J. Chen, Y. Chen, H. Ouyang and A. Luo, "Phase-Plane Analysis and Optimization of Dual Active Bridge DC Converters," ***Journal of Power Supply***, vol. 19, no. 2, pp. 81–92, Mar. 2021. (in Chinese)	

Journal Papers – Corresponding Author

No.	Publication Title	IF
[1]	Y. Zeng, **Z. Xiao**, Q. Liu, G. Liang, E. Rodriguez, G. Zou, X. Zhang, and J. Pou, "Physics-Informed Deep Transfer Reinforcement Learning Method for the Input-Series Output-Parallel Dual Active Bridge-Based Auxiliary Power Modules in Electrical Aircraft," ***IEEE Trans. Transport. Electrific.***, vol. 11, no. 2, pp. 6629–6639, Apr. 2025, doi: 10.1109/TTE.2024.3514657.	7.2
[2]	Z. Yao, X. He, **Z. Xiao**, F. Deng, C. Dai, S. Lu, and Y. Tang, "Current Ripple Prediction and ZVS-Based Variable Switching Frequency Control for Interleaved Multiphase Three-Level DC–DC Converter," ***IEEE Trans. Power Electron.***, vol. 40, no. 4, pp. 5109–5119, Apr. 2025, doi: 10.1109/TPEL.2024.3521997.	6.7
[3]	Z. Yao, L. Jiang, T. Sun, **Z. Xiao**, W. Chen and Y. Tang, "Smooth Optimal Trajectory and Voltage Balancing Control for T-Type Three-Level LLC Resonant Converter in Alternating Mode," ***IEEE Trans. Power Electron.***, vol. 40, no. 7, pp. 9278–9288, Jul. 2025, doi: 10.1109/TPEL.2025.3548174.	6.7

Journal Papers – Co-Author

No.	Publication Title	IF
[1]	Z. Yao, **Z. Xiao** et al., "Sensorless Simultaneous Self-Balance Mechanism of Voltage and Current Based on Near-CRM in Interleaved Three-Level DC–DC Converter," ***IEEE Trans. Power Electron.***, vol. 39, no. 2, pp. 2086–2099, Feb. 2024, doi: 10.1109/TPEL.2023.3330812.	6.7
[2]	M. Chen et al., "MagNet Challenge for Data-Driven Power Magnetics Modeling," ***IEEE Open J. Power Electron.***, vol. 6, pp. 883–898, 2025, doi: 10.1109/OJPEL.2024.3469916.	5.0
[3]	Z. Yao, X. He, H. Lan, H. Yang, **Z. Xiao** et al., "A Trapezoidal Current Mode to Reduce Peak Current in Near-CRM for Three-Level DC–DC Converter," ***IEEE Trans. Power Electron.***, vol. 39, no. 8, pp. 9446–9456, Aug. 2024, doi: 10.1109/TPEL.2024.3399216.	6.7
[4]	F. Deng, L. Zhang, **Z. Xiao**, Z. Yao and Y. Tang, "Decentralized Multilayer Master-Slave Control Strategy for Power Management in Autonomous DC Microgrid Clusters," ***IEEE Trans. Ind. Electron.***, vol. 72, no. 3, pp. 2712–2723, Mar. 2025, doi: 10.1109/TIE.2024.3443949.	7.7
[5]	F. Deng, L. Zhang, **Z. Xiao**, Z. Yao and Y. Tang, "Inertia Emulation in Droop-Based DC Microgrids With Equivalent Converter Impedance Reshaping," ***IEEE Trans. Power Electron.***, vol. 39, no. 11, pp. 15206–15216, Nov. 2024, doi: 10.1109/TPEL.2024.3437767.	6.7
[6]	R. Hou, Y. Liu, Z. He, Z. Li, M. Xiao, J. Guan, Y. Chen, J. Wang, **Z. Xiao**, and H. Wang, "Optimization and Design of LCC Resonant High-Voltage Wide Range Power Supply for Magnetrons," ***IEEE J. Emerg. Sel. Topics Power Electron.***, vol. 12, no. 4, pp. 3835–3847, Aug. 2024, doi: 10.1109/JESTPE.2024.3412800.	5.5

No.	Publication Title	IF
[7]	Y. Li, S. Wang, Y. Wu, Y. Jiang, **Z. Xiao** and Y. Tang, "Heterogeneous Integration of Isotropic and Anisotropic Magnetic Cores for Inductive Power Transfer," ***IEEE Trans. Power Electron.***, vol. 40, no. 2, pp. 3770–3784, Feb. 2025, doi: 10.1109/TPEL.2024.3485193.	6.7
[8]	Z. Yao, X. He, M. Liu, J. Liu, **Z. Xiao** and Y. Tang, "Fixed Switching Frequency Control Using Trapezoidal Current Mode to Achieve ZVS in Three-Level DC–DC Converters," ***IEEE Trans. Ind. Electron.***, vol. 72, no. 4, pp. 4175–4185, Apr. 2025, doi: 10.1109/TIE.2024.3454707.	7.7
[9]	H. Yang, F. Deng, L. Zhang, **Z. Xiao**, Z. Yao and Y. Tang, "Power System Inertia Enhancement Based on LVDC Microgrids," ***IEEE Trans. Ind. Electron.***, doi: 10.1109/TIE.2024.3508069.	7.7
[10]	Y. Wu, Y. Jiang, Y. Li, **Z. Xiao**, H. Yuan, X. Wang, and Y. Tang, "Precise Coil Inductance Prediction in WPT Systems: A Transfer Learning Approach," ***IEEE Trans. Transport. Electrific.***, vol. 11, no. 2, pp. 7083–7095, Apr. 2025, doi: 10.1109/TTE.2024.3524106.	7.2
[11]	T. Sun, **Z. Xiao**, F. Deng, Z. He, X. Pan and Y. Tang, "Single-Stage Multiport AC–DC–DC Converter With Reactive Power Control Capability," ***IEEE Trans. Power Electron.***, vol. 40, no. 5, pp. 6700–6713, May 2025, doi: 10.1109/TPEL.2025.3532439.	6.7
[12]	Z. Yao, **Z. Xiao**, et al., "Generalized Trapezoidal Current Mode-Based Zero-Voltage Switching for Multilevel DC–DC Converters," ***IEEE Trans. Power Electron.***, vol. 40, no. 7, pp. 8956–8961, Jul. 2025, doi: 10.1109/TPEL.2025.3551341.	6.7
[13]	G. Luo, Z. Yao, L. Jiang, **Z. Xiao**, W. Chen and Y. Tang, "Current Balancing Control Method With Reduced Number of Current Sensors for Interleaved LLC Converters," ***IEEE J. Emerg. Sel. Topics Power Electron.***, vol. 12, no. 3, pp. 2604–2613, Jun. 2024, doi: 10.1109/JESTPE.2024.3388481.	5.5
[14]	X. He, Z. Yao, X. Yang, B. B. Ngo, **Z. Xiao**, B. Wang, S. Lu and Y. Tang, "Hybrid TZCM–QCM Control Method for Smooth Power Transition and ZVS in a Bidirectional Three-Level DC-DC Converter," ***IEEE Trans. Power Electron.***, vol. 41, no. 4, pp. 6328–6338, 2026, doi: 10.1109/TPEL.2025.3630091.	6.7
[15]	X. He, Z. Yao, **Z. Xiao**, B. Yang, Y. Chen, Q. Xu, W. Chen and S. Lu, "A Unified Trapezoidal Current Mode for Symmetric Three-Level Buck-Boost Converters Without Mode Switching," ***IEEE Trans. Power Electron.***, 2026.	6.7

Conference Papers – First Author

No.	Publication Title	P.S.
[1]	**Z. Xiao**, Z. Yao, Y. Zeng and Y. Tang, "Light Load Audible-Noise-Free Skip Control for Dual Active Bridge Converters," ***IEEE Int. Conf. Power Electron. Drive Syst. (PEDS)***, Penang, Malaysia, 2025, pp. 1–5, doi: 10.1109/PEDS63958.2025.11144783.	
[2]	**Z. Xiao**, L. Zhang, F. Deng and Y. Tang, "Comparisons of Cycle-by-Cycle Start-Up Control in Synchronous Boost Converters," ***IEEE Int. Conf. Power Electron. Drive Syst. (PEDS)***, Penang, Malaysia, 2025, pp. 1–5, doi: 10.1109/PEDS63958.2025.11144889.	
[3]	**Z. Xiao**, Y. Jiang, C. Wang and Y. Tang, "A Streamlined and Intuitive AI-Powered Analysis and Design Tool for Resonant Converters," ***IEEE Int. Power Electron. Motion Control Conf. (IPEMC-ECCE Asia)***, Chengdu, China, 2024, pp. 852–857.	**Best Paper Award First Prize 3/927**
[4]	**Z. Xiao**, Y. Jiang, Z. Yao and Y. Tang, "AI-assisted Peak Efficiency Searching of CLLC Resonant Converters in Step-down Conditions," ***IEEE Int. Power Electron. Motion Control Conf. (IPEMC-ECCE Asia)***, Chengdu, China, 2024, pp. 863–868.	
[5]	**Z. Xiao**, L. Zhang, Z. He and Y. Tang, "Efficiency Improvement of DAB Converters with Enhanced Inductor Current Classification," ***IEEE Energy Convers. Congr. Expo. (ECCE)***, Nashville, TN, USA, 2023, pp. 2631–2638.	
[6]	**Z. Xiao**, Z. He, F. Deng and Y. Tang, "Power Loss Minimization of CLLC Resonant Converters via Time Domain Analysis," ***IEEE Energy Convers. Congr. Expo. (ECCE)***, Nashville, TN, USA, 2023, pp. 3392–3399.	
[7]	**Z. Xiao**, Y. Jiang, Y. Jiang and Y. Tang, "A Zero-Knowledge ANN-Based Waveform and Critical Parameter Calculator for Resonant Converters," ***Annu. Conf. IEEE Ind. Electron. Soc. (IECON)***, Singapore, 2023, pp. 1–6.	
[8]	**Z. Xiao**, Y. Jiang, Z. Yao and Y. Tang, "A Precise and Fast BPNN-Based Voltage Gain Model of CLLC Converters in All Operation Conditions," ***Annu. Conf. IEEE Ind. Electron. Soc. (IECON)***, Singapore, 2023, pp. 1–6.	
[9]	**Z. Xiao**, Z. He, A. Luo, Q. Xu, X. Huang and J. Zhang, "A Modular High-Frequency Resonant DC-DC Transformer for MVDC Application," ***IEEE Int. Power Electron. Appl. Conf. Expo. (PEAC)***, Shenzhen, China, 2018, pp. 1–6.	

Conference Papers – Co-Author

No.	Publication Title	P.S.
[1]	D. Zhao, Y. Wu, Y. Li, R. Zhang, **Z. Xiao** and Y. Tang, "A Convenient Harmonic State Space Model for ZVS Analysis in Wireless Power Transfer Systems," ***IEEE Energy Convers. Congr. Expo. (ECCE)***, Philadelphia, PA, USA, 2025, pp. 1–6, doi: 10.1109/ECCE58356.2025.11259601.	
[2]	Y. Zeng, J. Pou, G. Zou, H. Jie, **Z. Xiao**, et al., "Physics-Informed Neural Network for DC Solid State Transformers in DC Microgrids," ***Annu. Conf. IEEE Ind. Electron. Soc. (IECON)***, Madrid, Spain, 2025, pp. 1–6, doi: 10.1109/IECON58223.2025.11221331.	

No.	Publication Title	P.S.
[3]	Z. Yao, J. Liu, S. Yao, B. Ngo, **Z. Xiao** and Y. Tang, "Exploring the Soft-Switching Benefits of TZCM Mode in Three-Level DC-DC Converters Using Wide Bandgap Power Devices," ***IEEE Workshop Wide Bandgap Power Devices Appl. Asia (WiPDA Asia)***, Beijing, China, 2025, pp. 1–5, doi: 10.1109/WiPDA-Asia63772.2025.11183701.	
[4]	Z. Yao, J. Liu, F. Deng, **Z. Xiao**, Y. Jiang and Y. Tang, "A Five-Level Buck Converter with Zero-Voltage Switching Based on Trapezoidal Current Mode," ***IEEE Int. Conf. Power Electron. Drive Syst. (PEDS)***, Penang, Malaysia, 2025, pp. 1–5, doi: 10.1109/PEDS63958.2025.11144861.	
[5]	Z. Yao, X. He, **Z. Xiao**, F. Deng, W. Chen and Y. Tang, "A Novel Trapezoidal Current Mode for Realizing Zero-Voltage Switching in Three-Level DC-DC Converters," ***IEEE Int. Power Electron. Motion Control Conf. (IPEMC-ECCE Asia)***, Chengdu, China, 2024, pp. 4894–4899.	
[6]	F. Deng, S. Zhang, L. Zhang, Z. Yao, **Z. Xiao** and Y. Tang, "Decentralized Current Sharing in Islanded DC Microgrids Based on AC Frequency Droop," ***IEEE Int. Power Electron. Motion Control Conf. (IPEMC-ECCE Asia)***, Chengdu, China, 2024, pp. 1364–1369.	
[7]	L. Zhang, **Z. Xiao**, H. Wang, Y. Tang, S. Yang and X. Qi, "Analyze and Mitigate Low-Frequency Ripple of Arm Current for Modular Multilevel Converter based Solid-State Transformers," ***IEEE Int. Power Electron. Motion Control Conf. (IPEMC-ECCE Asia)***, Chengdu, China, 2024, pp. 1108–1113.	
[8]	L. Zhang, **Z. Xiao**, F. Deng, Z. Yao, H. Yang and Y. Tang, "Grid-Side High-Order Harmonics Mitigation for Three-Phase Four-Wire Inverter Based on Repetitive Control and Virtual Filter," ***IEEE Int. Power Electron. Motion Control Conf. (IPEMC-ECCE Asia)***, Chengdu, China, 2024, pp. 1079–1083.	
[9]	Z. Yao, X. He, **Z. Xiao**, Z. He, Y. Jiang and Y. Tang, "Full-Range ZVS Control Method Based on Near-CRM for T-Type Three-Level Inverter," ***IEEE Energy Convers. Congr. Expo. (ECCE)***, Nashville, TN, USA, 2023, pp. 2987–2992.	
[10]	Y. Jiang, Y. Wu, Y. Li, Z. He, **Z. Xiao**, H. Wang, X. Wang and Y. Tang, "Ultra-high Efficiency Wireless Power Transfer Systems Based on Novel Multi-variable Control Strategy," ***IEEE Energy Convers. Congr. Expo. (ECCE)***, Nashville, TN, USA, 2023, pp. 1659–1665.	
[11]	Y. Wu, Y. Jiang, Y. Li, Z. He, C. Wang, **Z. Xiao**, X. Wang and Y. Tang, "A Universal Coil Structure Design Method with Accurate Numerical Computation in Wireless Power Transfer Systems," ***IEEE Energy Convers. Congr. Expo. (ECCE)***, Nashville, TN, USA, 2023, pp. 6497–6504.	
[12]	Y. Jiang, Y. Wu, Y. Li, **Z. Xiao**, N. Wang, M. Wu, X. Wang and Y. Tang, "Precise Modeling for the Inductance of Rounded Rectangular Coils in Wireless Power Transfer Systems," ***Annu. Conf. IEEE Ind. Electron. Soc. (IECON)***, Singapore, 2023, pp. 1–7.	
[13]	T. Sun, **Z. Xiao**, Z. He and Y. Tang, "Single-Stage Isolated Three-Port AC-DC-DC Converter with Asymmetrical Modulation," ***IEEE Appl. Power Electron. Conf. Expo. (APEC)***, Long Beach, CA, USA, 2024, pp. 501–505.	
[14]	Z. He, H. Ding, Z. Zhang, J. Wu, D. Zhang, J. Shao, **Z. Xiao**, Z. Yao, and Y. Tang, "A Dynamic Sampling Method to Achieve Noise-Free Sampling for High-Frequency Converters," ***IEEE Appl. Power Electron. Conf. Expo. (APEC)***, Long Beach, CA, USA, 2024, pp. 2789–2793.	
[15]	L. Zhang, H. Yang, **Z. Xiao**, Z. Yao, F. Deng, T. Sun, and Y. Tang, "Capacitor Voltage-Balancing Method of Three-Phase Four-Wire T-Type Inverter Based Active Power Filter," ***IEEE Appl. Power Electron. Conf. Expo. (APEC)***, Long Beach, CA, USA, 2024, pp. 2277–2281.	
[16]	Y. Li, S. Wang, Y. Wu, Y. Jiang, X. Zhu, **Z. Xiao**, Z. He, and Y. Tang, "A Novel Hybrid Magnetic Core Design Method for Weight Reduction of Wireless Power Transfer Systems," ***IEEE Appl. Power Electron. Conf. Expo. (APEC)***, Long Beach, CA, USA, 2024, pp. 1202–1206.	
[17]	Z. He, H. Ding, Z. Zhang, J. Wu, D. Zhang, Y. Z. Liu, **Z. Xiao**, Z. Yao, and Y. Tang, "Coordinate Control of NP Voltage Balance in Two-Stage Three-level DC-AC Converters," ***IEEE Appl. Power Electron. Conf. Expo. (APEC)***, Long Beach, CA, USA, 2024, pp. 2749–2755.	
[18]	F. Deng, H. Wang, **Z. Xiao**, L. Zhang, Z. Yao and Y. Tang, "Decentralized Master-Slave Control Strategy for Current Sharing in Islanded DC Microgrids," ***IEEE South. Power Electron. Conf. (SPEC)***, Brisbane, Australia, 2024, pp. 1–5, doi: 10.1109/SPEC62217.2024.10893277.	
[19]	B. Huang, Y. Zeng, D. Zhou, G. Cheng, **Z. Xiao**, X. Han, J. Zou and J. Pou, "A Novel State of Health Estimation Framework for Lithium Iron Phosphate Batteries in DC Microgrids," ***IEEE Int. Conf. DC Microgrids (ICDCM)***, 2026, pp. 1–5.	
[20]	B. B. Ngo, Z. Yao, **Z. Xiao**, J. Liu and Y. Tang, "Efficiency Enhancement of Three-Level Flying Capacitor Totem-Pole PFC with Trapezoidal Current Mode Control," ***IEEE Appl. Power Electron. Conf. Expo. (APEC)***, 2026, pp. 84–88.	

Patents Filed

No.	Patent Title
1	A. Luo, **Z. Xiao** et al., *SWITCHED CAPACITOR RESONANT VOLTAGE-MULTIPLYING RECTIFIER CONVERTER, AND A CONTROL METHOD AND A CONTROL SYSTEM THEREOF*. WO2023005270A1.
2	Z. He, **Z. Xiao** et al., *Automatic voltage balance switch network, DC converter, control system and control method*. ZL202011485672.8.
3	A. Luo, **Z. Xiao** et al., *Switched capacitor resonance voltage doubler rectifier converter and its control method and control system*. ZL202110852407.7.
4	R. Guan, A. Luo, . . . **Z. Xiao** et al., *Modular combined medium-voltage DC converter and control method based on two-stage conversion structure*. L201910868861.4.
5	A. Luo, **Z. Xiao** et al., *Resonant DC converter with series input and multi-port output and its control method*. ZL201811235162.8.

No.	Patent Title
6	Z. He, **Z. Xiao** et al., *Voltage equalizing capacitor regulator for input series structure and its control method*. ZL202110849483.2.
7	Z. He, **Z. Xiao** et al., *A DC-DC conversion system and its control method*. ZL202110842999.4.
8	Z. He, B. Zhou, . . . **Z. Xiao** et al., *Submarine DC converter and control method*. ZL202010499215.8.
9	A. Luo, Z. He, . . . **Z. Xiao** et al., *A high-reliability and high-power case-based medium and high voltage DC power supply*. ZL202010499215.8.

Awards & Honors

Awards & Honors Title	Time
Presidential Scholarship for Doctoral Student of Hunan University 5/279	2020
Outstanding Papers of 2021 in "Journal of Power Supply" 3/334	2022
Session chair for the 49th Annual Conference of the IEEE Industrial Electronics Society	2023
Best paper award (First prize) in 2024 IPEMC Conference 3/927	2024
Poster Session Chair of 2024 2nd Power Electronics and Power System Conference	2024
Special Session Panelists for the International Decentralized Energy Access Solutions Conference	2025
Special Session Organizer for the 15th IEEE International Conference on Power Electronics and Drive Systems	2025
Technical Program Committee for 2025 IEEE Transportation Electrification Conference and Expo, Asia-Pacific (ITEC-AP)	2025
Special Issue Guest Editor for *Energies* Journal "Control and Optimization of Power Converters"	2025
Special Issue Guest Editor for *Electronics* Journal "Power Converters and Multilevel Converters"	2025

Teaching

Teaching Course Title	Time
NTU EEE EE6506 Semiconductor Based Converter in Renewable Energy Systems	2023 Fall
NTU EEE EE6506 Semiconductor Based Converter in Renewable Energy Systems	2024 Fall
NTU EEE EE6506 Semiconductor Based Converter in Renewable Energy Systems	2025 Fall
NTU EEE EE6501 Power Electronic Converters	2025 Fall
NTU EEE EE6514 Special Topics in Clean Energy System Design	2025 Fall
NTU EEE EE6501 Power Electronic Converters	2026 Spring
NTU EEE EE6501 Power Electronic Converters	2026 Fall

Students Mentoring / Co-Mentoring

Name	Class	Thesis Project Title
PAN TIANQI	M.SC.(POWER ENG.) Batch 2025	M.SC. | Ultra-Compact and High-Efficiency DC–DC Converters for Next-Generation Power Electronics
XU WEIZHONG	NG Batch 2025	FYP | Gate Driver Design for Next-Generation Wideband Gap Power Devices
MOU XINGYAN	M.SC.(POWER ENG.) Batch 2025	M.SC. | High-Efficiency High-Frequency Power Converter Design for Next-Generation AI Processors
DUAN YILIN	M.SC.(POWER ENG.) Batch 2024	EE6008 Collaborative Research and Development Project AY25S1-6-PE | High-Efficiency Power Converter Using Wide Bandgap Devices
MA QINGHAN	M.SC.(POWER ENG.) Batch 2024	EE6008 Collaborative Research and Development Project AY25S1-6-PE | High-Efficiency Power Converter Using Wide Bandgap Devices
CHEN KA YI	M.SC.(POWER ENG.) Batch 2025	EE6008 Collaborative Research and Development Project AY25S1-6-PE | High-Efficiency Power Converter Using Wide Bandgap Devices
LIU SHIQI	M.SC.(POWER ENG.) Batch 2025	EE6008 Collaborative Research and Development Project AY25S1-6-PE | High-Efficiency Power Converter Using Wide Bandgap Devices
ZHOU BEICHEN	M.SC.(POWER ENG.) Batch 2024	EE6008 Collaborative Research and Development Project AY25S1-6-PE | High-Efficiency Power Converter Using Wide Bandgap Devices
ZHAO WENPU	M.SC.(POWER ENG.) Batch 2025	M.SC. | AI-Assisted Power Loss Modelling and Efficiency Optimization for Power Converters
CHEN JUNYU	EEE Batch 2024	URECA EEE25040 | Optimizing Power Converter Design and Efficiency Through Transfer Learning

Name	Class	Thesis Project Title
HE SONGRUI	EEE Batch 2024	URECA EEE25041 | Ultra-Efficient and Ultra-Compact Power Supply for AI Chips
EDWIN TAN YONG HAO	EEE Batch 2023	FYP | Gate Driver Circuit Design for Wide Band Gap Power Devices
HU HAORAN	NG Batch 2024	FYP | High-Efficiency and High-Density Power Converter Design for AI Chips in Data Centres
ZHANG HUAYU	NG Batch 2024	FYP | Gate Driver Circuit Design for Wide Band Gap Power Devices
SIDHANTH NODU SANTHOSH	EEE Batch 2021	FYP | Gate Driver Circuit Design for Wide Band Gap Power Devices
IAN QUEK YONG EN	EEE Batch 2021	FYP | High-Efficiency and High-Density Power Converter Design for AI Chips in Data Centres
PANG YI HONG	EEE Batch 2023	URECA EEE24041 | Optimizing Power Converter Design and Efficiency Through Transfer Learning
CHERYL LYNN KOSASIH	EEE Batch 2021	FYP A1102-241 | AI-Assisted Design and Control of Next-Generation Power Electronic Converter for Enhanced Energy Efficiency and Power Density in Renewable Energy Systems
CUI QILONG	M.SC.(POWER ENG.) Batch 2024	M.SC. | Soft Transition of CLLC Resonant Converters for Ultra-Wide Range Battery Charging Applications
ZHANG YUNPENG	M.SC.(POWER ENG.) Batch 2024	M.SC. | Gate Driver Circuit Design for Wide Band Gap Power Devices
NGUYEN THANH TUNG	M.SC.(POWER ENG.) Batch 2024	EE6008 Collaborative Research and Development Project AY24S1-50 | Machine Learning - Assisted Power Loss Modelling and Efficiency Optimization for Power Converters
YUNCHENG	M.SC.(POWER ENG.) Batch 2024	EE6008 Collaborative Research and Development Project AY24S1-50 | Machine Learning - Assisted Power Loss Modelling and Efficiency Optimization for Power Converters
WEN YUHENG	M.SC.(POWER ENG.) Batch 2024	EE6008 Collaborative Research and Development Project AY24S1-50 | Machine Learning - Assisted Power Loss Modelling and Efficiency Optimization for Power Converters
DENG MIAO	M.SC.(POWER ENG.) Batch 2024	EE6008 Collaborative Research and Development Project AY24S1-50 | Machine Learning - Assisted Power Loss Modelling and Efficiency Optimization for Power Converters
YEO YEE JIE	EEE Batch 2021	FYP A1115-231 | Inductor Design for Buck-Boost Converter
HUANG YIYUAN	M.SC.(POWER ENG.) Batch 2023	M.SC. | Design and Control for a Bidirectional Interlinking Converter
XU DONGCHEN	M.SC.(POWER ENG.) Batch 2023	M.SC. | Optimal Inductor Design for a 6.6kW Buck-Boost Converter
LU ZHIYAO	M.SC.(POWER ENG.) Batch 2023	M.SC. | 20 kW SiC Bidirectional DC-DC Converter Design
LEI YIHAN	M.SC.(POWER ENG.) Batch 2022	M.SC. | Dynamic Response Improvement of a Bidirectional Interleaved Buck-Boost Converter
JIN SHUHAN	M.SC.(POWER ENG.) Batch 2022	M.SC. | Efficiency Optimization of an Interleaved Buck-Boost Converter with Wide Operation Range